

PH 521: Electromagnetic Theory

Course Code	PH 521
Course Name	Electromagnetic Theory
L-T-P-C	3-0-0-3
Prerequisites	Undergraduate core physics courses
Intended for	Core for M.Sc. and I-Ph.D. in Physics; Elective for B.Tech. 3rd and 4th Years.
Mutual Exclusion	NA
Approval	10th Senate (Jun 2016)
Revised	64th BoA (May 2026) - credits changed to 3-0-0-3

Preamble

This course will develop the concepts of electricity and magnetism at a level suitable for master's students. The course content includes electrostatics and magnetostatics and their unification into electrodynamics, and electromagnetic radiation. The idea of special theory of relativity has been included in classical electrodynamics.

Course Modules

- 1. **Electrostatics and Magnetostatics in Vacuum:** Review of electrostatics and magnetostatics, Laplace and Poisson equations, Boundary Value Problems in Electrostatics, Method of Images, Electrostatic and magnetostatic energy. (7 lectures)
- 2. **Electrostatics and Magnetostatics in Matter:** Multipole Expansions, Dielectrics and Magnetization, Field inside cavity, Clausius-Mossotti equation. (7 lectures)
- 3. **Maxwell equations:** Maxwell Equations, gauge symmetry, Electromagnetic energy and momentum, Conservation Laws. (6 lectures)
- 4. **Electromagnetic Waves:** Waves in one dimension, reflection, refraction and polarization, Electromagnetic waves in vacuum, Electromagnetic Waves in dispersive media, Absorption and Dispersion, Wave-guides. (11 lectures)
- 5. **EM Radiation:** Retarded Potential, Lienard-Wiechart potential, Radiation from accelerated charge, Dipole radiation, Larmor formula. (7 lectures)
- 6. **Special Relativity and Electromagnetism:** Magnetism as a relativistic phenomenon, Lorentz transformations of electromagnetic field, Lorentz covariance of Maxwell equations. (4 lectures)

Text Books

1. D.J. Griffiths, *Introduction to Electrodynamics* 5th ed, CUP (2025)

References

1. J.D. Jackson, *Classical Electrodynamics* 3rd ed, Wiley (2007)
2. A. Zangwill, *Modern Electrodynamics*, CUP (2012)
3. K. Milton and J. Schwinger, *Classical Electrodynamics*, CRC Press (1998)
4. M. Maggiore, *A Modern Introduction to Classical Electrodynamics*, OUP (2023)